Chest X-Ray Pneumonia Detection — Capstone Report

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1. Approach

This project builds an automated pneumonia screening system from pediatric chest X-rays, using the public Chest X-Ray Images (Pneumonia) dataset. The training split is notably imbalanced (~3:1 PNEUMONIA:NORMAL), which is the central technical risk of the task: a naive model can score high on plain accuracy while being clinically useless if it simply favors the majority class. Two models were built and compared under identical preprocessing: a baseline CNN trained from scratch (four convolutional blocks), and a ResNet18 fine-tuned via transfer learning (ImageNet-pretrained backbone, with the final residual block and classifier head unfrozen for fine-tuning). The class imbalance was addressed at two levels during training: a WeightedRandomSampler rebalanced each training batch, and the loss function used class weights, so the minority (NORMAL) class was not drowned out during optimization.

2. Results

Metric	Baseline CNN	ResNet18 (Transfer Learning)
Accuracy	0.8317	0.9071
Precision	0.8740	0.8915
Recall	0.8538	0.9692
F1	0.8638	0.9287
ROC-AUC	0.8930	0.9573

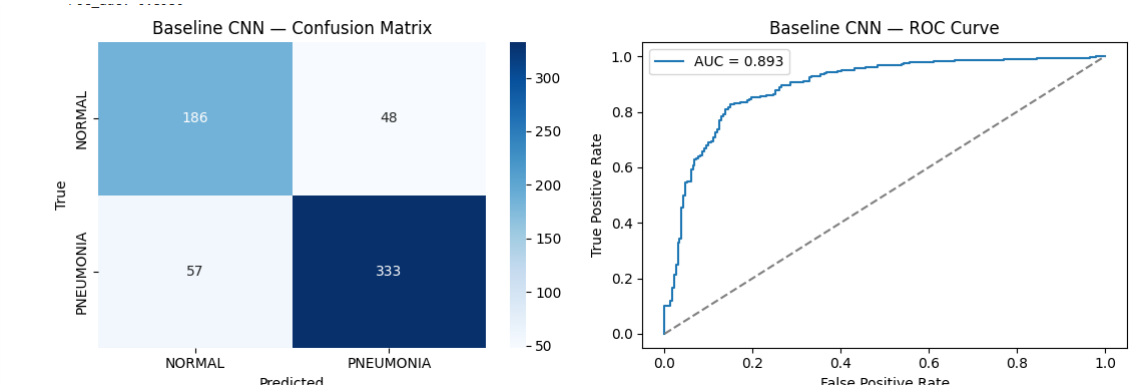


Figure 1. Baseline CNN — confusion matrix and ROC curve (test set).

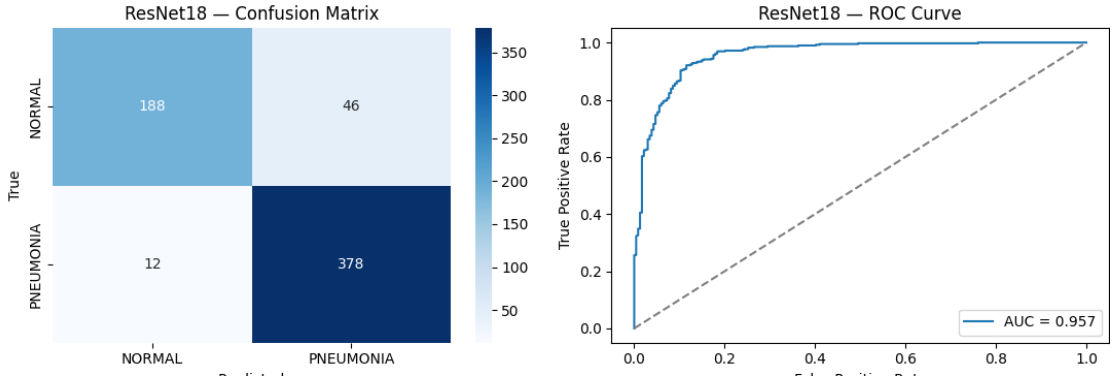


Figure 2. ResNet18 (transfer learning) — confusion matrix and ROC curve (test set).

Both models perform reasonably, but the baseline CNN's recall (0.85) trails ResNet18's (0.97) by a wide margin — the confusion matrix shows the baseline misses 57 real pneumonia cases out of 390, versus only 12 for ResNet18. The ResNet18 model is more balanced overall, with F1 of 0.93 versus 0.86 for the baseline, and a higher ROC-AUC (0.957 versus 0.893), while keeping false positives on the NORMAL class similarly low (46 versus 48).

3. Chosen Model & Why

Chosen model: ResNet18 (transfer learning).

Accuracy alone was not used to select the model, because the ~3:1 class imbalance makes accuracy an unreliable signal — a model that always predicted PNEUMONIA would still score a misleadingly high accuracy without being clinically useful. Instead, the decision followed the course's model-selection framework by weighing precision, recall, F1, and ROC-AUC together:

- **F1** balances precision and recall, and ResNet18's F1 (0.93) is clearly above the baseline's (0.86) — driven mainly by ResNet18's stronger recall (0.97 vs 0.85).
- **ROC-AUC**, a threshold-independent measure, also favors ResNet18 (0.957 vs 0.893), confirming the improvement isn't just an artifact of the default 0.5 decision threshold.
- In a pneumonia-screening context, **recall matters more than precision**: a missed pneumonia case (false negative) is more costly than a false alarm (false positive), since a false positive typically just triggers a follow-up review. The confusion matrices make this concrete: the baseline CNN misses 57 real pneumonia cases out of 390 on the test set, while ResNet18 misses only 12. ResNet18's much higher recall makes it the clearly safer and better-justified choice, even though its precision is only marginally higher than the baseline's.

4. Known Limitations

- Dataset is pediatric chest X-rays only — may not generalize to adult patients.
- Single public dataset from one source/hospital — possible site-specific artifacts (e.g., scan markers, positioning conventions) the model could be partly relying on rather than purely lung pathology.

- Class imbalance was mitigated but not eliminated; ResNet18's confusion matrix shows 46 NORMAL images misclassified as PNEUMONIA on the test set (precision of 0.89, not 1.0), which should be monitored in any deployment.
- No external/held-out validation from a different hospital system, so real-world generalization is unverified.
- Some errors are made with very high confidence rather than being borderline cases (see Section 5) — high-confidence errors are harder to catch with a simple confidence threshold than low-confidence ones.

5. Live Demo Notes

During the live demo, 2-3 test images were sent to the running /predict endpoint and their returned label and confidence score were shown alongside the true label. The demo also included at least one misclassified example, deliberately located by scanning the test set for a case where the model's prediction disagreed with the true label.

Misclassification pattern: ResNet18's confusion matrix (Figure 2) shows 46 NORMAL images misclassified as PNEUMONIA on the test set, out of 234 total NORMAL images — this is the model's dominant error mode, more common than missing real PNEUMONIA cases (only 12 misses out of 390). Likely explanations:

- The misclassified images may contain visual artifacts (positioning, exposure, scan markers, or benign lung markings) that resemble the texture patterns the model associates with pneumonia (e.g., localized opacity or haziness).
- Because the model was trained with a weighted sampler to compensate for the PNEUMONIA-heavy training set, it may have become slightly biased toward predicting PNEUMONIA on ambiguous or atypical NORMAL images — the same correction that boosted recall likely raised the false-positive rate somewhat on the NORMAL class, which matches the model's precision of 0.89 (not 1.0).
- This is exactly the kind of failure a Grad-CAM visualization is designed to diagnose: overlaying which regions of the X-ray drove the prediction would show whether the model focused on lung tissue or on an irrelevant image region.